BEE 4750 Homework 4: Linear Programming and Capacity Expansion

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Due Date

Thursday, 11/06/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to analytically (or graphically) solve a linear program.
- Problem 2 asks you to formulate and solve a resource allocation problem using linear programming.
- Problem 3 asks you to formulate, solve, and analyze a standard generating capacity expansion problem.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
  Activating project at `~/BEE 4750/hw4-parker-regina`
```

```julia
using JuMP
using HiGHS
using DataFrames
using Plots
using Measures
using CSV
using MarkdownTables
```

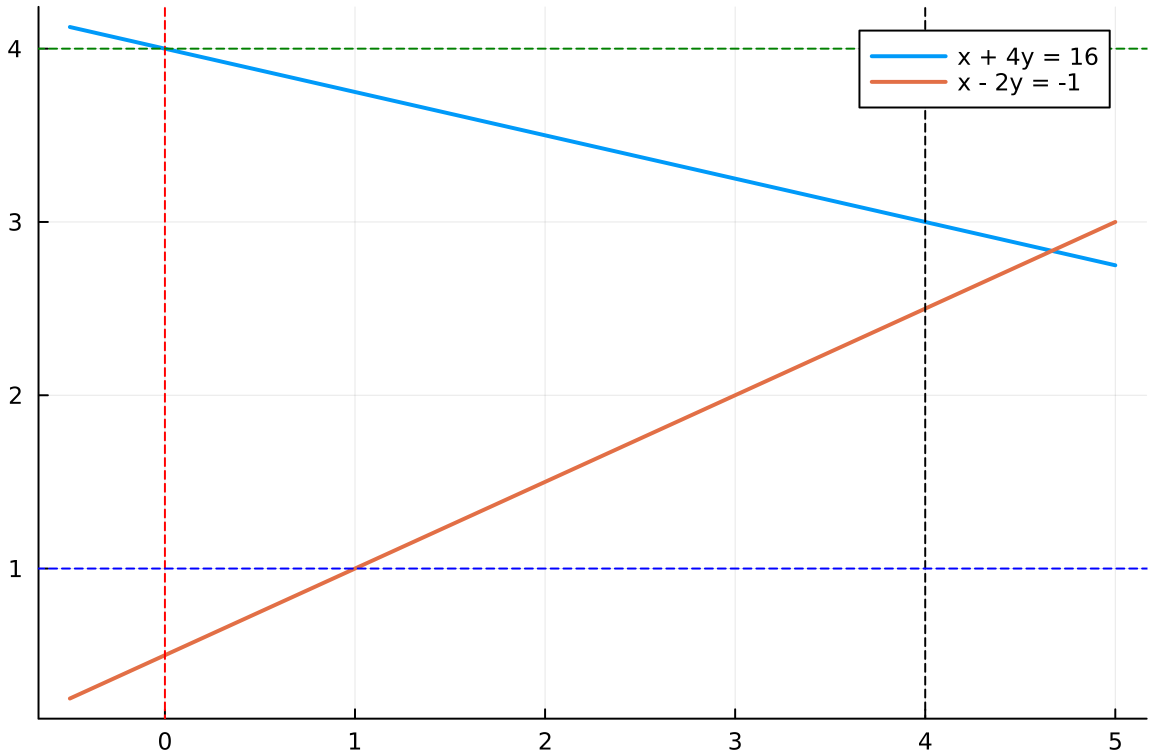
Problems (Total: 30 Points)

Problem 1 (Total: 3 Points)

Solve the following linear program **analytically**. Show your work and justify any reasoning.

$$\begin{aligned} \max_{x,y} \quad & 3x + 2y \\ \text{subject to:} \quad & x + 4y \leq 16 \\ & x - 2y \leq -1 \\ & 0 \leq x \leq 4 \\ & 1 \leq y \leq 4. \end{aligned}$$

```julia
#Set x range for plotting
x = -0.5:0.01:5.0

#Constraint functions
y1(x) = (16 - x) / 4
y2(x) = (x + 1) / 2

#Plot constraint functions across all x values
plt = plot(x, y1.(x), lw=2, label="x + 4y = 16")
plot!(plt, x, y2.(x), lw=2, label="x - 2y = -1")

#Draw the vertical and horizontal lines representing
#the bounds for constraints
vline!([0], linestyle=:dash, color=:red, label="")
vline!([4], linestyle=:dash, color=:black, label="")
hline!([1], linestyle=:dash, color=:blue, label="")
hline!([4], linestyle=:dash, color=:green, label="")
```



Using the above graph of constraints, we know that the maximum value must occur at an intersection of the boundaries. We can use these intersections to check the values for the maximum of the function $3x+2y$. The feasible intersections, as determined by are as follows:

$$(0,1),(0,4),(1,1),(4,\frac{5}{2}),(4,3)$$

We can then check these pairs with the function $3x+2y$:

$$3(0)+2(1)=2, \quad (0,1)$$

$$3(0)+2(4)=8, \quad (0,4)$$

$$3(1)+2(1)=5, \quad (1,1)$$

$$3(4)+2(2.5)=17, \quad (4,2.5)$$

$$3(4) + 2(3) = 18, \quad (4, 3)$$

Therefore, we can see that at $(4, 3)$ we achieve the maximum value of $3x + 2y = 18$.

Problem 2 (12 points)

A farmer has access to a pesticide which can be used on corn, soybeans, and wheat fields, and costs \$70/kg-yr to apply. The crop yields the farmer can obtain following crop yields by applying varying rates of pesticides to the field are shown in Table 1.

Application Rate (kg/ha)	Soybean (kg/ha)	Wheat (kg/ha)	Corn (kg/ha)
0	2900	3500	5900
1	3800	4100	6700
2	4400	4200	7900

Table 1: Crop yields from applying varying pesticide rates for Problem 1.

The costs of production, *excluding pesticides*, for each crop, and selling prices, are shown in Table 2.

Crop	Production Cost (\$/ha-yr)	Selling Price (\$/kg)
Soybeans	350	0.36
Wheat	280	0.27
Corn	390	0.22

Table 2: Costs of crop production, excluding pesticides, and selling prices for each crop.

Recently, environmental authorities have declared that farms cannot have an *average* application rate on soybeans, wheat, and corn which exceeds 0.8, 0.7, and 0.6 kg/ha, respectively. The farmer has asked you for advice on how they should plant crops and apply pesticides to maximize profits over 130 total ha while remaining in regulatory compliance if demand for each crop (which is the maximum the market would buy) this year is 250,000 kg?

Problem 2.1

Formulate a linear program for this resource allocation problem, including clear definitions of decision variable(s) (including units), objective function(s), and constraint(s) (make sure to explain functions and constraints with any needed derivations and explanations). **Tip: Make sure that all of your constraints are linear**.

SOLUTION 2.1 To formulate a linear program for this resource allocation problem, we started off by defining nine decision variables: three for each crop of the crops representing the number of hectares allocated to the pesticide application rate of each crop.

$$s_r, w_r, c_r \text{ for } r = 0, 1, 2$$

Each of these decision variables satisfy the non-negativity constraint:

$$s_r,\ w_r,\ c_r \geq 0 \quad \text{for } r = 0, 1, 2$$

The goal is to maximize profit so the objective function follows the equation: \$ profit = revenue - cost \$.

$$\max Z = \sum_{r=0}^{2}(p_s y_{s,r} s_r + p_w y_{w,r} w_r + p_c y_{c,r} c_r) - \left(C_s \sum_{r=0}^{2} s_r + C_w \sum_{r=0}^{2} w_r + C_c \sum_{r=0}^{2} c_r \right) - 70 \sum_{r=0}^{2}(r s_r + r w_r + r c_r)$$

There are a few constraints included in this LP. First of all, the farm only has 130 ha so the total amount of land used has to be less than or equal to 130 ha.

$$\sum_{r=0}^{2} s_r + \sum_{r=0}^{2} w_r + \sum_{r=0}^{2} c_r \leq 130$$

Second, each crop has a demand of $250,000kg$ and the production of each crop is capped at this demand limit.

$$\sum_{r=0}^{2} y_{s,r} s_r \leq 250000$$

$$\sum_{r=0}^{2} y_{w,r} w_r \leq 250000$$

$$\sum_{r=0}^{2} y_{c,r} c_r \leq 250000$$

Finally, each crop there is a regulatory limit for each crop on the average pesticide application rate. To find the average pesticide application rate, we used the following equations:

$$\frac{0s_0 + 1s_1 + 2s_2}{s_0 + s_1 + s_2} \leq 0.8$$

$$\frac{0w_0 + 1w_1 + 2w_2}{w_0 + w_1 + w_2} \leq 0.7$$

$$\frac{0c_0 + 1c_1 + 2c_2}{c_0 + c_1 + c_2} \leq 0.6$$

However, all the constraints must be linear so the equations were manipulated to:

$$0s_0 + 1s_1 + 2s_2 \leq 0.8(s_0 + s_1 + s_2)$$

$$0w_0 + 1w_1 + 2w_2 \leq 0.7(w_0 + w_1 + w_2)$$

$$0c_0 + 1c_1 + 2c_2 \leq 0.6(c_0 + c_1 + c_2)$$

Problem 2.2

How many ha should the farmer dedicate to each crop and with what pesticide application rate(s)? How much profit will the farmer expect to make?

SOLUTION 2.2 The linear program recommends:

Soybeans: 69.06 ha total

13.81 ha at 0 kg/ha

55.25 ha at 1 kg/ha

Wheat: 22.48 ha total

6.74 ha at 0 kg/ha

15.73 ha at 1 kg/ha

Corn: 38.46 ha total

26.92 ha at 0 kg/ha

11.54 ha at 2 kg/ha

This will yield a maximized profit of $ 116,741 while remaining within the land constraint of 130 ha limit.

To summarize, to maximize profit the farmer should plant approximately 69.06 ha of soybeans primarily using application rate 1, 22.48 ha of wheat also mostly using application rate 1, and 38.46 ha of corn using a mix of rate 0 and rate 2 applications.

Problem 2.3

The farmer has an opportunity to buy an extra 10 ha of land. How much extra profit would this land be worth to the farmer? Discuss why this value makes sense and under what conditions you would recommend the farmer should make the purchase.

SOLUTION 2.3

The shadow price of the land constraint is $ 729.40 / ha. This represents the increase in profit if one additional hectare of land were available to grow crops on.

So, the value of an additional 10 hectares of land is:

$ 10 * 729.40 = $ 7,294 $

According to the results from the LP, soybeans and corn have both hit the 250,000 kg demand limit. This means that the farmer could only plan wheat on the 10 extra ha of land since that is the only crop that has not hit the demand limit. From our findings explained in 2.2 we know that for wheat we will apply $\frac{6.74}{22.48} * 100 = 30\%$ at rate 0 and $\frac{15.74}{22.48} * 100 = 70\%$ at rate 1 and so the yield is $ 0.3 * 3500 + 0.7 * 4100 = 3920$ kg/ha. Hence, the revenue is $0.27 * 3920$ = $ $1,058.40$ / ha,the production cost is $ 280 / ha, and the pesticide cost is $70 * 0.7 =$ $ 49 / ha. This results in the profit per extra ha to be:

$1,058.40 - 280 - 49 =$ $ 729.40 / extra ha

This is exactly the shadow price of land so this value of $ $7,294$ makes sense!

The farmer should make the purchase if the annualized cost of buying 10 hectares is less than $ $7,294$.

```
#Problem 2.1
using JuMP # optimization modeling syntax
using HiGHS # optimization solver

crop_model = Model(HiGHS.Optimizer) # initialize model object
@variable(crop_model, s[0:2] >= 0) # non-neg constraints for soy
@variable(crop_model, w[0:2] >= 0) # non-neg constraitns for wheat
@variable(crop_model, c[0:2] >= 0) # non-neg constraitns for corn
```

```
# index: 1 = soy, 2 = wheat, 3 = corn

# set up variables:
# selling prices ($/kg)
ps, pw, pc = 0.36, 0.27, 0.22
# production cost ($/ha)
cs, cw, cc = 350, 280, 390
# pesticide cost ($ per kg/ha applied)
pest = 70
# yields (kg/ha)
ys0, ys1, ys2 = 2900, 3800, 4400
yw0, yw1, yw2 = 3500, 4100, 4200
yc0, yc1, yc2 = 5900, 6700, 7900
# average rate caps (kg/ha)
Ls, Lw, Lc = 0.8, 0.7, 0.6
# land & demand
land_cap = 130
demand  = 250000 #for each crop

# OBJ: maximize profits over 130 total ha while remaining in regulatory
# compliance if demand for each crop (which is the maximum the market
# would buy) this year is 250,000 kg
# Objective = revenue - production costs - pesticide costs
@objective(crop_model, Max,
    # revenue
    + ps*(ys0*s[0] + ys1*s[1] + ys2*s[2])
    + pw*(yw0*w[0] + yw1*w[1] + yw2*w[2])
    + pc*(yc0*c[0] + yc1*c[1] + yc2*c[2])
    # production costs
    - cs*sum(s[i] for i in 0:2)
    - cw*sum(w[i] for i in 0:2)
    - cc*sum(c[i] for i in 0:2)
    # pesticide costs (rate * pest * hectares)
    - pest*( 0*s[0] + 1*s[1] + 2*s[2]
           + 0*w[0] + 1*w[1] + 2*w[2]
           + 0*c[0] + 1*c[1] + 2*c[2])
)

# Constraints

# Land
@constraint(crop_model, land, sum(s[i] for i in 0:2) + sum(w[i] for i in 0:2) + sum(c[i] for
```

```
# Demand caps (kg)
@constraint(crop_model, ds, ys0*s[0] + ys1*s[1] + ys2*s[2] <= demand)   # soy
@constraint(crop_model, dw, yw0*w[0] + yw1*w[1] + yw2*w[2] <= demand)   # wheat
@constraint(crop_model, dc, yc0*c[0] + yc1*c[1] + yc2*c[2] <= demand)   # corn

# Average pesticide caps: sum(rate*x)   L * sum(x)
@constraint(crop_model, pest_s, (0*s[0] + 1*s[1] + 2*s[2]) <= Ls*(sum(s[i] for i in 0:2)))
@constraint(crop_model, pest_w, (0*w[0] + 1*w[1] + 2*w[2]) <= Lw*(sum(w[i] for i in 0:2)))
@constraint(crop_model, pest_c, (0*c[0] + 1*c[1] + 2*c[2]) <= Lc*(sum(c[i] for i in 0:2)))  #


print(crop_model)
```

```
Max 694 s[0] + 948 s[1] + 1094 s[2] + 665.0000000000001 w[0] + 757 w[1] + 714 w[2] + 908 c[0
Subject to
 land : s[0] + s[1] + s[2] + w[0] + w[1] + w[2] + c[0] + c[1] + c[2]   130
 ds : 2900 s[0] + 3800 s[1] + 4400 s[2]   250000
 dw : 3500 w[0] + 4100 w[1] + 4200 w[2]   250000
 dc : 5900 c[0] + 6700 c[1] + 7900 c[2]   250000
 pest_s : -0.8 s[0] + 0.19999999999999996 s[1] + 1.2 s[2]   0
 pest_w : -0.7 w[0] + 0.30000000000000004 w[1] + 1.3 w[2]   0
 pest_c : -0.6 c[0] + 0.4 c[1] + 1.4 c[2]   0
 s[0]   0
 s[1]   0
 s[2]   0
 w[0]   0
 w[1]   0
 w[2]   0
 c[0]   0
 c[1]   0
 c[2]   0
```

```
optimize!(crop_model);

println("Optimized soybean values:")
@show value.(s[i] for i in 0:2);

println("Optimized wheat values:")
@show value.(w[i] for i in 0:2);

println("Optimized corn values:")
```

```julia
@show value.(c[i] for i in 0:2);

println("Optimized objective value:")
@show objective_value(crop_model);
```

```
Running HiGHS 1.11.0 (git hash: 364c83a51e): Copyright (c) 2025 HiGHS under MIT licence terms
LP   has 7 rows; 9 cols; 27 nonzeros
Coefficient ranges:
  Matrix [2e-01, 8e+03]
  Cost   [7e+02, 1e+03]
  Bound  [0e+00, 0e+00]
  RHS    [1e+02, 2e+05]
Presolving model
7 rows, 9 cols, 27 nonzeros  0s
7 rows, 9 cols, 27 nonzeros  0s
Presolve : Reductions: rows 7(-0); columns 9(-0); elements 27(-0) - Not reduced
Problem not reduced by presolve: solving the LP
Using EKK dual simplex solver - serial
  Iteration        Objective     Infeasibilities num(sum)
          0    -1.8170297740e+02 Ph1: 7(28.3601); Du: 9(181.703) 0s
          7     1.1674116702e+05 Pr: 0(0) 0s
Model status        : Optimal
Simplex   iterations: 7
Objective value     :  1.1674116702e+05
P-D objective error :  0.0000000000e+00
HiGHS run time      :          0.00
Optimized soybean values:
value.((s[i] for i = 0:2)) = [13.812154696132604, 55.24861878453038, 0.0]
Optimized wheat values:
value.((w[i] for i = 0:2)) = [6.743306417339563, 15.734381640458977, 0.0]
Optimized corn values:
value.((c[i] for i = 0:2)) = [26.92307692307692, 0.0, 11.538461538461547]
Optimized objective value:
objective_value(crop_model) = 116741.16702082448
```

```julia
println("Shadow values:")
@show shadow_price(land);

@show shadow_price(ds);
@show shadow_price(dw);
@show shadow_price(dc);
```

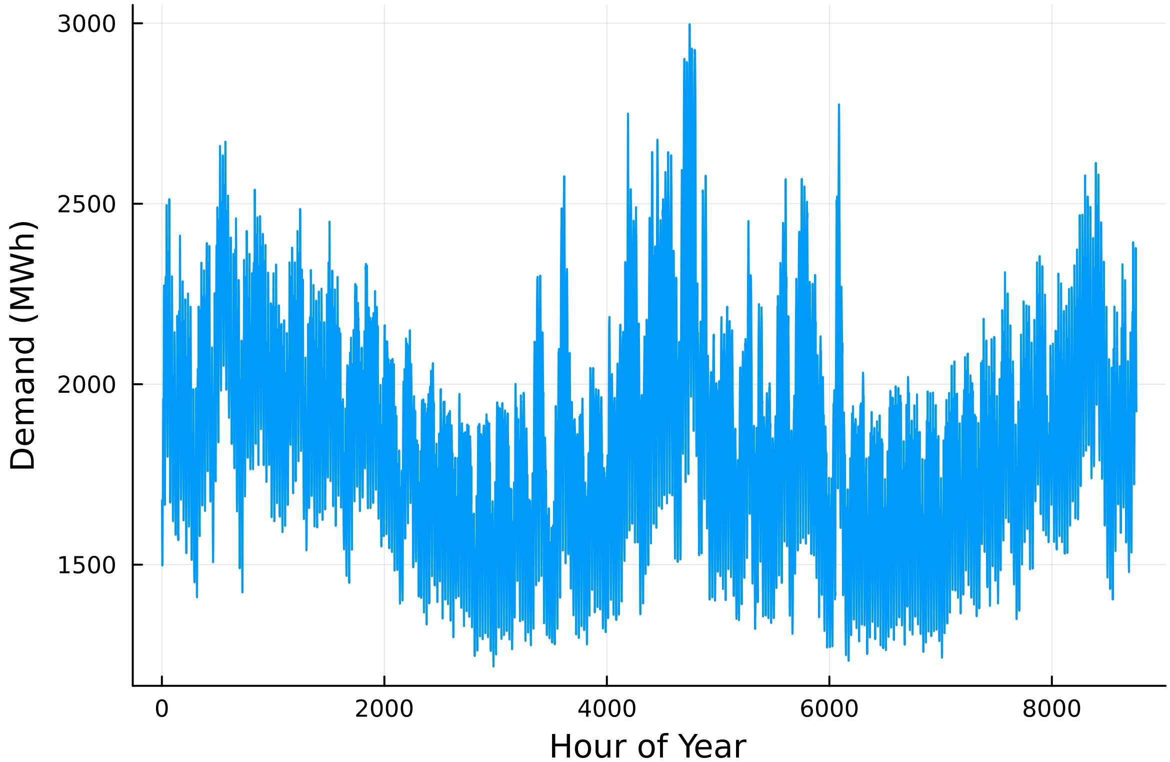
```julia
@show shadow_price(pest_w);
@show shadow_price(pest_w);
@show shadow_price(pest_c);
```

```
Shadow values:
shadow_price(land) = 729.4
shadow_price(ds) = 0.046353591160220996
shadow_price(dw) = -0.0
shadow_price(dc) = 0.04132307692307693
shadow_price(pest_w) = 91.99999999999993
shadow_price(pest_w) = 91.99999999999993
shadow_price(pest_c) = 108.67692307692309
```

Problem 3 (15 points)

For this problem, we will use hourly load (demand) data from 2013 in New York's Zone C (which includes Ithaca). The load data is loaded and plotted below in Figure 1.

```julia
# load the data, pull Zone C, and reformat the DataFrame
NY_demand = DataFrame(CSV.File("data/2013_hourly_load_NY.csv"))
rename!(NY_demand, :"Time Stamp" => :Date)
demand = NY_demand[:, [:Date, :C]]
rename!(demand, :C => :Demand)
demand[:, :Hour] = 1:nrow(demand)

# plot demand
plot(demand.Hour, demand.Demand, xlabel="Hour of Year", ylabel="Demand (MWh)", label=:false)
```



Next, we load the generator data, shown in Table 3. This data includes fixed costs ($/MW installed), variable costs ($/MWh generated), and CO2 emissions intensity (tCO2/MWh generated).

```
gens = DataFrame(CSV.File("data/generators.csv"))
```

	Plant	FixedCost	VarCost	Emissions
	String15	Int64	Int64	Float64
1	Geothermal	450000	0	0.0
2	Coal	220000	24	1.0
3	NG CCGT	82000	30	0.43
4	NG CT	65000	40	0.55
5	Wind	91000	0	0.0
6	Solar	70000	0	0.0

Finally, we load the hourly solar and wind capacity factors, which are plotted in Figure 2. These tell us the fraction of installed capacity which is expected to be available in a given hour for generation (typically based on the average meteorology).

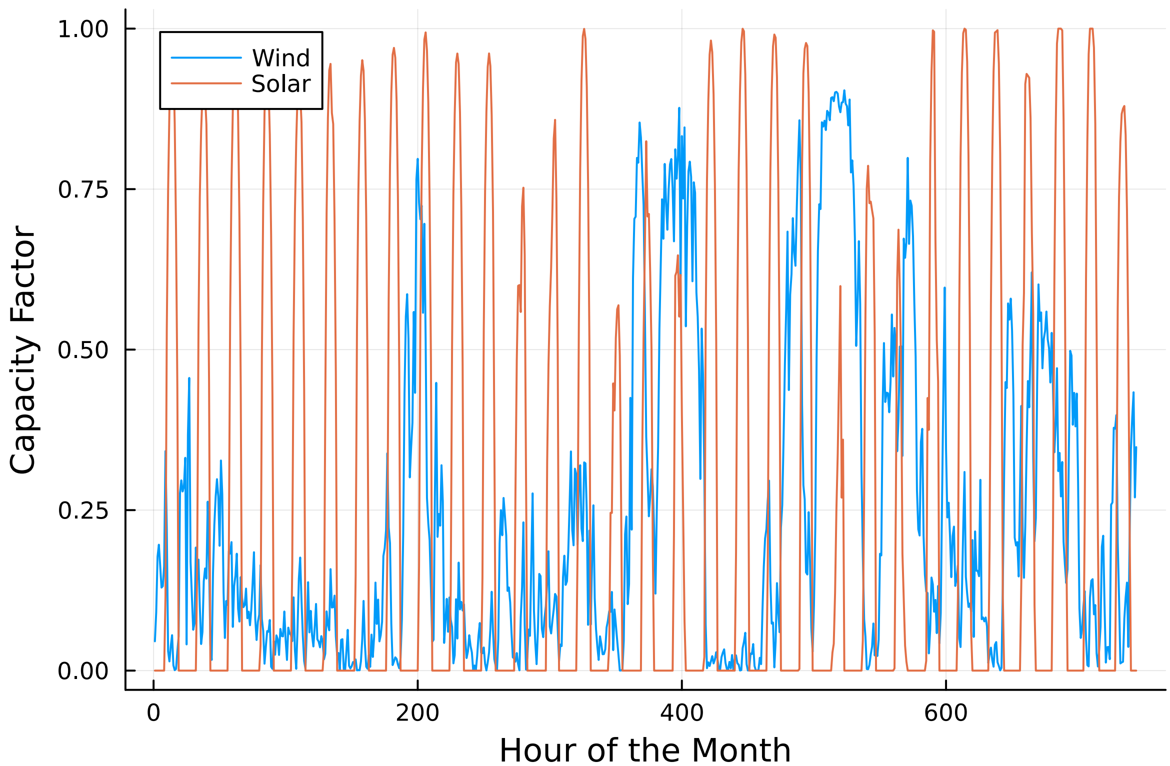
```
# load capacify factors into a DataFrame
cap_factor = DataFrame(CSV.File("data/wind_solar_capacity_factors.csv"))

# plot January capacity factors
p1 = plot(cap_factor.Wind[1:(24*31)], label="Wind")
plot!(cap_factor.Solar[1:(24*31)], label="Solar")
xaxis!("Hour of the Month")
yaxis!("Capacity Factor")

p2 = plot(cap_factor.Wind[4344:4344+(24*31)], label="Wind")
plot!(cap_factor.Solar[4344:4344+(24*31)], label="Solar")
xaxis!("Hour of the Month")
yaxis!("Capacity Factor")

display(p1)
display(p2)
```

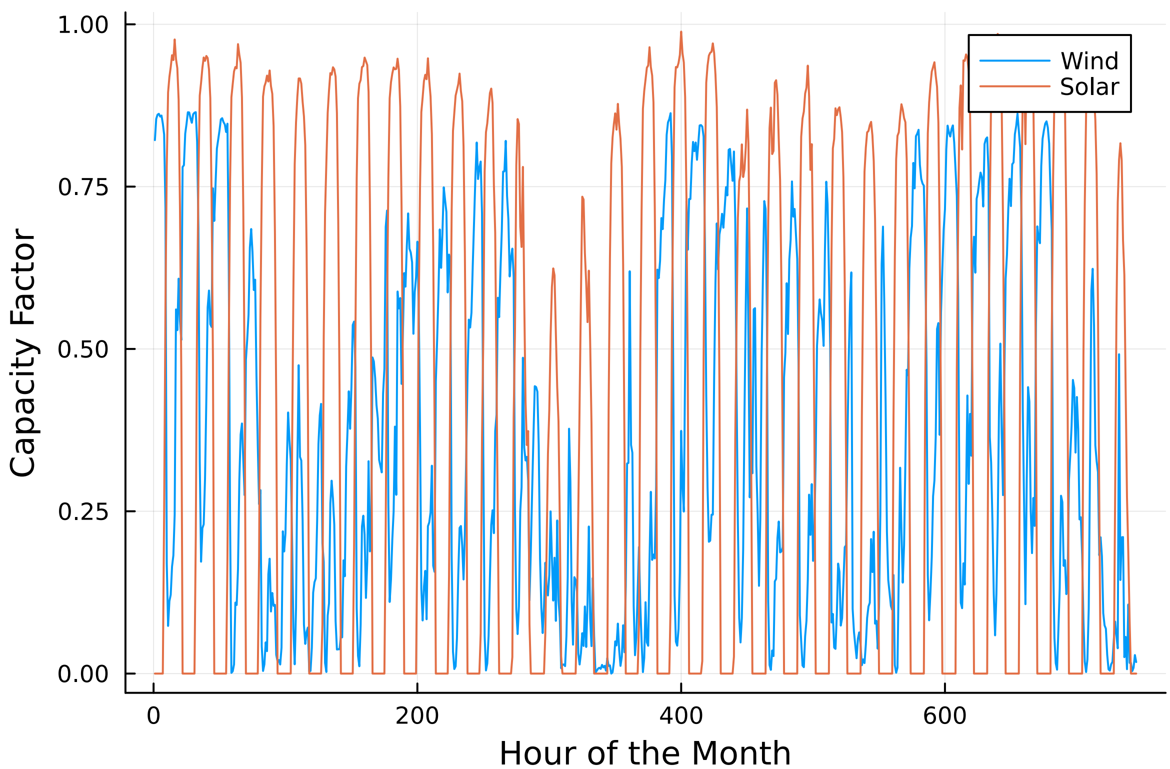
You have been asked to develop a generating capacity expansion plan for the utility in Riley County, NY, which currently has no existing electrical generation infrastructure. The utility can build any of the following plant types: geothermal, coal, natural gas combined cycle gas turbine (CCGT), natural gas combustion turbine (CT), solar, and wind. The NY state legislature is planning to enact an annual CO2 limit, which for the utility would limit the emissions in its footprint to 1.5 MtCO2/yr.

While coal, CCGT, and CT plants can generate at their full installed capacity, geothermal plants operate at maximum 85% capacity, and solar and wind available capacities vary by the hour depend on the expected meteorology. The utility will also penalize any non-served demand at a rate of $10,000/MWh.

Problem 3.1

Formulate a linear program for this capacity expansion problem, including clear definitions of decision variable(s) (including units), objective function(s), and constraint(s) (make sure to explain functions and constraints with any needed derivations and explanations).


Wind
Solar
Capacity Factor
1.00
0.75
0.50
0.25
0.00
0
200
400
600
Hour of the Month


Wind
Solar
Capacity Factor
1.00
0.75
0.50
0.25
0.00
0
200
400
600
Hour of the Month

$$
\begin{aligned}
\min_{x,y,NSE} \quad & \sum_{g\in\mathcal{G}} \text{FixedCost}_g \times x_g + \sum_{t\in\mathcal{T}}\sum_{g\in\mathcal{G}} \text{VarCost}_g \times y_{g,t} + \sum_{t\in\mathcal{T}} \text{NSECost} \times NSE_t \\
\text{subject to:} \quad & \sum_{g\in\mathcal{G}} y_{g,t} + NSE_t \geq d_t & \forall t \in \mathcal{T} \\
& y_{g,t} \leq \eta_{g,t} \times x_g & \forall g \in \mathcal{G}, \forall t \in \mathcal{T} \\
& \sum_{t\in\mathcal{T}}\sum_{g\in\mathcal{G}} \text{Emissions}_g \times y_{g,t} \leq TE & \forall g \in \mathcal{G}, \forall t \in \mathcal{T} \\
& x_g, y_{g,t}, NSE_t \geq 0 & \forall g \in \mathcal{G}, \forall t \in \mathcal{T}
\end{aligned} \tag{1}
$$

** Note these equations are the same as used in lecture except for Emissions_g and $\eta_{g,t}$

Decision Variables:

x_g : Installed capacity of generation technology g [MW]

$y_{g,t}$: Energy generated by plant g during time t [MWh]

NSE_t : Non-served energy during time period t [MWh]

Parameters:

FixedCost_g : Annualized fixed cost per MW of installed capacity for technology g [$/MW-yr]

VarCost_g : Variable operating cost of technology g [$/MWh]

NSECost : Penalty cost for non-served energy [$/MWh] = 10,000 $/MWh

d_t : Electricity demand over time t [MWh].

$\eta_{g,t}$: Capacity factor of technology g in time period t

$Coal, CCGT, CT : \eta_{g,t} = 1$

$Geothermal : \eta_{g,t} = 0.85$

$Solar, Wind : \eta_{g,t}$varies hourly based on meteorology

Emissions_g : CO_2 emissions rate of technology g [tCO2/MWh]

TE : Annual CO_2 emissions limit = 1.5 MtCO2/yr

$\mathcal{G}$: Set of generation technologies: {coal, CCGT, CT, geothermal, solar, wind}

$\mathcal{T}$: Total time scale

Plant [g]	FixedCost [$]	VarCost [$/MWh]	Emissions [tCO2/MWh]
Geothermal	450000	0	0.00

Plant [g]	FixedCost [$]	VarCost [$/MWh]	Emissions [tCO2/MWh]
Coal	220000	24	1.00
NG CCGT	82000	30	0.43
NG CT	65000	40	0.55
Wind	91000	0	0.00
Solar	70000	0	0.00

Problem 3.2

How much should the utility build of each type of generating plant? What will the total cost be? How much energy will be non-served?

Solution

Plant Type	Installed Capacity (MW)
Geothermal	285.568
Coal	0.0
NG CCGT	1509.17
NG CT	703.062
Wind	2548.89
Solar	2103.75

Total cost = $\$ \ 785,352,958.94$

Total unserved energy per year (MWh) = 332.29

Problem 3.3

What fraction of annual generation does each plant type produce? How does this compare to the breakdown of built capacity that you found in Problem 3.2? Do these results make sense given the demand and generator data?

Solution

Plant	Annual Output (MWh)	Fraction of Generation (%)
Geothermal	1.10949e6	6.78
Coal	0.0	0.0
NG CCGT	3.32277e6	20.30
NG CT	1.29473e5	0.79
Wind	6.61471e6	40.41

Plant	Annual Output (MWh)	Fraction of Generation (%)
Solar	5.19114e6	31.72

This breakdown of annual output by plant type makes sense when looking at installed capacity. What we see is that wind and solar each generate the most power, followed by natural gas and CCGT geothermal, and lastly by natural gas CT. In both installed capacity and annual output, coal plants are not used which is likely do to the emissions capacity and high fixed cost, compared to the similar variable costs and lower emissions of natural gas CCGT. These results make logical sense when considering the results of Question 3.3, and observing the "gens" dataframe output."

Problem 3.4

Make a plot of the electricity price in each hour. Discuss any trends that you see.

Solution

The graph shows that wholesale electricity prices remain low and relatively stable for most hours of the year, generally tracking the variable cost of the marginal generator. But there are a few hours that clearly stand out because the price spikes sharply to $10,000/MWh.

This happens when the system is unable to meet demand with available capacity and the emissions cap, forcing the model to rely on unserved energy (NSE), which has a cost of $10,000/MWh. In these scarcity hours, the dual value of the load-balance constraint equals –10,000, which becomes a market-clearing price of $10,000/MWh. Economically, this indicates that an additional 1 MWh of supply during those hours would be worth $10,000 to the system, highlighting extreme scarcity conditions. The result shows that under the emissions limit and annual demand profile, the system is generally able to serve load at normal marginal costs, but a small number of peak hours create stress where additional capacity would significantly reduce system costs.

Problem 3.5

What is the shadow price of CO2? What is the value to the utility be of allowing it to emit an additional 1000 tCO2/yr?

Solution

The shadow price of being able to emit an additional $ 182.77 tCO2/yr, while the utility of being able to emit 1000 tCO2/yr is $\$\ 182,771.94$ per year.

Significant Digits

Use `round(x; digits=n)` to report values to the appropriate precision! If your number is on a different order of magnitude and you want to round to a certain number of significant digits, you can use `round(x; sigdigits=n)`.

Getting Variable Output Values

`value.(x)` will report the values of a `JuMP` variable `x`, but it will return a special container which holds other information about `x` that is useful for `JuMP`. This means that you can't use this output directly for further calculations. To just extract the values, use `value.(x).data`.

Suppressing Model Command Output

The output of specifying model components (variable or constraints) can be quite large for this problem because of the number of time periods. If you end a cell with an `@variable` or `@constraint` command, I *highly* recommend suppressing output by adding a semi-colon after the last command, or you might find that your notebook crashes.

```
#Question 3.2

#Vectorize lists to create workable variables
G = 1:nrow(gens)
T = 1:nrow(demand)
FixedCost = gens.FixedCost
VarCost = gens.VarCost
Emissions = gens.Emissions
d = demand.Demand

#Initialize costs
NSECost = 10000
TE = 1.5e6

# Build eta[g,t] which is the availability x capacity factor
eta = zeros(length(G), length(T))
#Iterate through each generator and assign capacity factors depending on plant type
for (g_idx, Plant) in enumerate(gens.Plant)
    if Plant in ["Coal", "NG CCGT", "NG CT"]
        eta[g_idx, :] .= 1.0
    elseif Plant == "Geothermal"
        eta[g_idx, :] .= 0.85
    elseif Plant == "Solar"
```

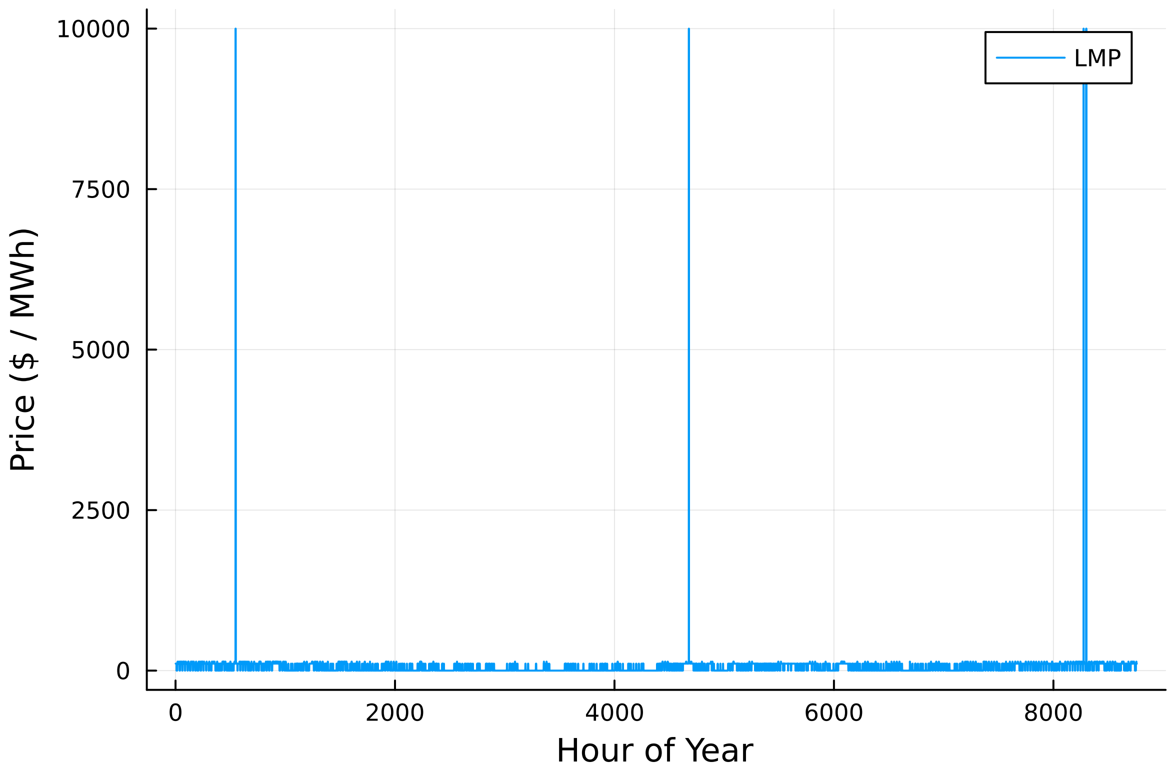
```julia
        eta[g_idx, :] .= cap_factor.Solar
    elseif Plant == "Wind"
        eta[g_idx, :] .= cap_factor.Wind
    end
end

gencap = Model(HiGHS.Optimizer)
@variables(gencap, begin
    x[g in G] >= 0
    y[g in G, t in T] >= 0
    NSE[t in T] >= 0
end)

@objective(gencap, Min,
    sum(FixedCost[g] * x[g] for g in G)
    +
    sum(VarCost[g] * sum(y[g,t] for t in T) for g in G)
    +
    NSECost * sum(NSE[t] for t in T)
)

#Energy demand balance for each hour
@constraint(gencap, load[t in T], sum(y[g,t] for g in G) + NSE[t] >= d[t])

#Generation availability fits within installed capacity times the capacity factor
@constraint(gencap, availability[g in G, t in T], y[g,t] <= eta[g,t] * x[g])

#Maximum emissions
@constraint(gencap, emissions_cap, sum(Emissions[g] * y[g,t] for g in G, t in T) <= TE)
optimize!(gencap)


### Output results

#Find optimum values
x_opt = value.(x).data
y_opt = value.(y).data
NSE_opt = value.(NSE).data

#Question 3.2
#Capacity by plant [MW]
cap_dist = DataFrame(Plant = gens.Plant, Capacity_MW = [x_opt[g] for g in G])
```

```
show(cap_dist, allrows=true, allcols=true)

#Total annual cost [$ per year]
total_cost = objective_value(gencap)
println("Total cost per year = ", total_cost)

#Total annual unserved energy [MWh/year]
tot_NSE = sum(NSE_opt[t] for t in T)
println("Total unserved energy (MWh) = ", tot_NSE)


#Question 3.3
#Amount of annual generation by plant and share of total generation
plant_out = [sum(y_opt[g,t] for t in T) for g in G]
tot_gen = sum(plant_out)
gen_dist = plant_out ./ tot_gen
gen_fraction_summary = DataFrame(
    Plant = gens.Plant,
    Annual_MWh = plant_out,
    Percent_gen = 100 * gen_dist)

show(gen_fraction_summary, allrows=true, allcols=true)

#Question 3.4
# plot of electricity price in each hour
using Statistics
using Plots

# Duals (shadow prices) of hourly load constraints
dual_load = [shadow_price(load[t]) for t in T]      # these are   0 when binding
price = [-dual_load[t] for t in T]                  # $/MWh (economic interpretation)

#Show Prices
@show minimum(price), mean(price), maximum(price)

#Identify any scarcity hours (with NSE) and confirm price hits the penalty ($10,000/MWh)
scarcity_hours = findall(t -> value(NSE[t]) > 1e-6, T)
@show length(scarcity_hours)
@show unique(round.(price[scarcity_hours]; digits=2))[1:min(end,5)]  # should be ~10000.00

#Plot: Hourly wholesale electricity price over the year
p = plot(
```

```
    1:length(T), price,
    xlabel = "Hour of Year",
    ylabel = "Price (\$ / MWh)",
    label = "LMP",
    legend = :topright,
)
display(p)

#Question 3.5
#Shadow price of additional tCO2/yr
shadow = shadow_price(emissions_cap) * -1
println("Shadow price of additional tCO2/yr = ", shadow, " per year")
shadow_price_mt = shadow * 1000
println("Shadow price of additional 1000 tCO2/yr = ", shadow_price_mt, " per year")
```

```
Running HiGHS 1.11.0 (git hash: 364c83a51e): Copyright (c) 2025 HiGHS under MIT licence terms
LP   has 61321 rows; 61326 cols; 188256 nonzeros
Coefficient ranges:
  Matrix [5e-05, 1e+00]
  Cost   [2e+01, 4e+05]
  Bound  [0e+00, 0e+00]
  RHS    [1e+03, 2e+06]
Presolving model
56857 rows, 56862 cols, 179328 nonzeros  0s
56854 rows, 56859 cols, 179322 nonzeros  0s
Presolve : Reductions: rows 56854(-4467); columns 56859(-4467); elements 179322(-8934)
Solving the presolved LP
Using EKK dual simplex solver - serial
  Iteration        Objective     Infeasibilities num(sum)
          0     0.0000000000e+00 Pr: 8760(4.11193e+06) 0s
      28308     7.5146782481e+08 Pr: 3538(635046); Du: 0(3.55585e-06) 5s
      36200     7.8535295894e+08 Pr: 0(0); Du: 0(2.3938e-13) 10s
Solving the original LP from the solution after postsolve
Model status        : Optimal
Simplex   iterations: 36200
Objective value     :  7.8535295894e+08
P-D objective error :  1.3357583179e-14
HiGHS run time      :          9.87
6×2 DataFrame
 Row   Plant       Capacity_MW
       String15    Float64
```

```
   1   Geothermal      285.568
   2   Coal              0.0
   3   NG CCGT        1509.17
   4   NG CT           703.062
   5   Wind           2548.89
   6   Solar          2103.75Total cost per year = 7.853529589434903e8
Total unserved energy (MWh) = 332.2907137925944
6×3 DataFrame
 Row   Plant       Annual_MWh  Percent_gen
       String15    Float64     Float64

   1   Geothermal   1.10949e6     6.77857
   2   Coal         0.0           0.0
   3   NG CCGT      3.32277e6    20.3009
   4   NG CT        1.29473e5     0.791036
   5   Wind         6.61471e6    40.4135
   6   Solar        5.19114e6    31.716(minimum(price), mean(price), maximum(price)) = (-0.0,
length(scarcity_hours) = 6
(unique(round.(price[scarcity_hours]; digits = 2)))[1:min(end, 5)] = [10000.0]
Shadow price of additional tCO2/yr = 182.77193609185863 per year
Shadow price of additional 1000 tCO2/yr = 182771.93609185863 per year
```



References

List any external references consulted, including classmates.